

Main results

Positive Summation Kernels

A family of functions $(K_n)_{n \geq 1}$ on $\mathbb{R}$ (or on $\mathbb{T}$, the circle) is called a *positive summation kernel* if the following hold:

(i) $K_n(x) \geq 0$ for all $x \in \mathbb{R}$ and each n ;

(ii) $\int_{-\infty}^{\infty} K_n(x) dx = 1$ for all n ;

(iii) for every $\delta > 0$,

$$\int_{|x| \geq \delta} K_n(x) dx \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Equivalently, (K_n) is an *approximate identity*: the mass of K_n concentrates near $x = 0$ as $n \rightarrow \infty$. For such kernels, the convolution $(f * K_n)(x) = \int f(x - t)K_n(t) dt$ satisfies

$$(f * K_n)(x) \rightarrow f(x)$$

for all x where f is continuous.

Riemann-Lebesgue Lemma

Theorem 2.2 (Riemann–Lebesgue lemma) Let f be absolutely integrable in the Riemann sense on a finite or infinite interval I . Then

$$\lim_{\lambda \rightarrow \infty} \int_I f(u) \sin(\lambda u) du = 0.$$

(A corresponding result holds for $\cos(\lambda u)$ or $e^{i\lambda u}$.)

Corollary for Fourier Coefficients

Corollary. Let $f \in L^1(\mathbb{T})$ (i.e., $\int_{-\pi}^{\pi} |f(x)| dx < \infty$). Its complex Fourier coefficients are:

$$\hat{f}(n) = c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx, \quad n \in \mathbb{Z}.$$

The Fourier coefficients of any integrable function vanish at infinity:

$$\hat{f}(n) \longrightarrow 0 \quad \text{as } |n| \rightarrow \infty.$$

Proof Sketch of Corollary: The Fourier coefficient $\hat{f}(n)$ can be expressed as:

$$\hat{f}(n) = \frac{1}{2\pi} \left(\int_{-\pi}^{\pi} f(x) \cos(nx) dx + i \int_{-\pi}^{\pi} f(x) \sin(-nx) dx \right)$$

Since $\cos(nx)$ and $\sin(nx)$ are bounded functions, the absolute integrability of f is preserved when multiplied by these functions. As $|n| \rightarrow \infty$, both the real and imaginary parts of the integral tend to zero by applying the Riemann–Lebesgue lemma (and its cosine variant) with $\lambda = n$. Therefore, $\hat{f}(n) \rightarrow 0$ as $|n| \rightarrow \infty$.